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FINAL PROJECT REPORT

CS2311.CH190: Research topics on selected problem in computer science

INTEGRATING SOFT ACTOR-CRITIC WITH QUALITY-DIVERSITY FOR HIGH-DIMENSIONAL CONTINUOUS CONTROL

Team member (Team 9):

Nguyễn Thái Huy – 240101011 – huynt.19@grad.uit.edu.vn

Bùi Đăng Phúc – 240101064 – phucbd.19@grad.uit.edu.vn

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ABSTRACT

Learning robust and adaptable control policies in high-dimensional continuous environments remains a central challenge in reinforcement learning (RL). Traditional RL and evolutionary algorithms typically optimize for a single high-performing agent, which often lacks generalizability when facing small variations in reward functions or environment dynamics. Quality-Diversity (QD) algorithms address this limitation by evolving a archive of diverse and competent agents, enabling adaptability and robustness across tasks. Recent advances, such as Descriptor-Conditioned Reinforcement Learning MAP-Elites (DCRL-ME), incorporate RL into the QD framework and condition policies on behavioral descriptors, thereby encouraging specialization across niches. However, existing approaches typically rely on Twin-Delayed Deep Deterministic Policy Gradient (TD3) as the RL backbone, which struggles to scale in high-dimensional continuous control tasks due to instability and sensitivity to local optima.

In this work, we propose DCRL-ME (SAC), a novel variant that replaces TD3 with the Soft Actor-Critic (SAC) algorithm. By reformulating the descriptor-conditioned policy as a stochastic Gaussian policy and incorporating entropy-regularized optimization, our method improves both robustness and exploration. Experimental results on three Brax benchmarks (HalfCheetah-Uni, Ant-Omni, and Humanoid-Uni) demonstrate that DCRL-ME (SAC) achieves consistently higher QD Scores and Max Fitness than TD3-based variants. Notably, in the challenging Humanoid-Uni environment, our method achieves up to an eight-fold improvement in QD Score. Additionally, to address SAC's sensitivity to reward scaling, we conduct extensive evaluations across different scaling hyperparameters to provide intuitive knowledge of how to tune them.

I. INTRODUCTION

In many real-world control problems, such as robotics, autonomous navigation, or resource management, agents must continuously interact with their environments in order to learn how to act effectively. These agents gather experiences through trial and error, gradually refining their behavior based on the outcomes they observe. The standard formulation of this process is to optimize the agent's policy so that it maximizes the expected cumulative reward over time for a given task in a specific environment. Reinforcement learning (RL) [29] has become the dominant paradigm for tackling this objective, with evolutionary algorithms (EAs) [26] often providing an alternative optimization strategy. Both approaches, however, typically share one important limitation: they tend to return *a single high-performing agent* that is over-specialized for the particular environment and task used during training, hindering the generalization capabilities [25]. Even slight modifications in the reward structure or small perturbations in the environment's transition dynamics can cause a dramatic drop in performance [1]. For example, a robot trained to walk efficiently on a smooth surface may fail completely if the ground becomes slippery, or if one of its joints is damaged.

Interestingly, biological evolution may provide some hints on how to solve this problem. Instead, it does not produce a single best organism but instead preserves a wide spectrum of organisms that are specialized in different ways, each adapted to thrive under particular circumstances. When the environment changes, some organisms are likely to survive and succeed because their traits happen to align with the new conditions. Inspired by this a family of algorithms known as Quality-Diversity (QD) [10, 25] has been developed. Rather than seeking one globally optimal solution, QD methods aim to discover *a collection of diverse, high-quality agents*, each excelling in a particular behavioral niche. As illustrated in Figure 1, QD algorithms tend to find a spectrum of behaviors according to some features (or descriptors). This enables useful gaits that can be later used to adapt to different

environments or tasks. Traditionally, QD algorithms operate through evolutionary processes that rely on standard variation operators, such as crossover and random mutation. While effective, these mechanisms can be slow to converge, especially compared to the sample efficiency of modern RL methods [22]. Recognizing this limitation, recent research has begun to integrate RL into the QD framework, using RL as a more directed and powerful mutation mechanism [22, 33]. This hybridization accelerates the discovery of diverse and competent agents, combining the strengths of RL's fast learning with QD's ability to maintain diversity. The synergy between QD and RL represents an exciting and rapidly growing research direction. By balancing quality (high-performing solutions) and diversity (coverage of many behavioral strategies), these approaches are not only effective in training agents in their original environment but also resilient to perturbations, adaptable to new contexts, and capable of generalization.

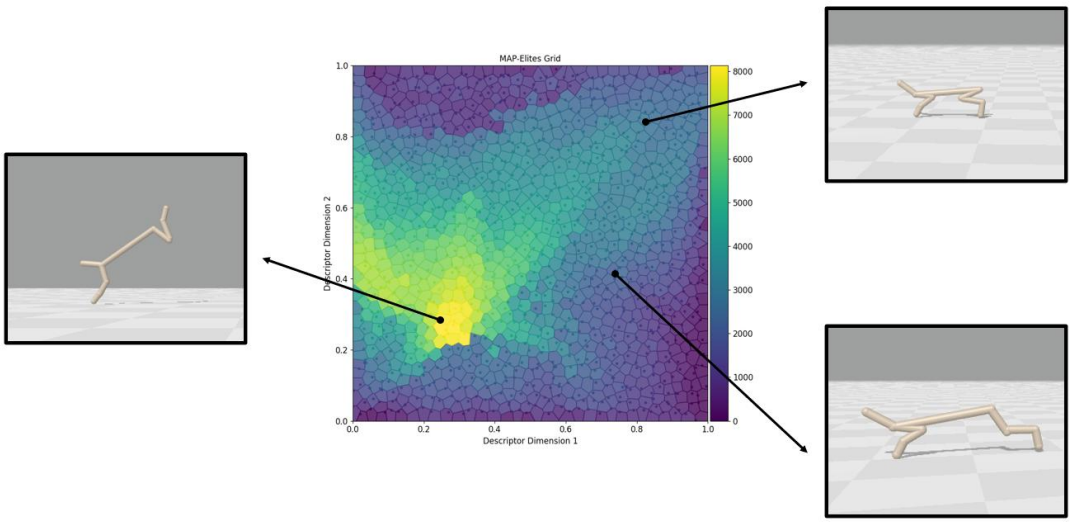


Figure 1: *Illustration of diverse behaviors from a Quality-Diversity algorithm.*

However, naively combining QD with RL may result in some disavantages. Conventional RL algorithms are designed to optimize the quality of agents by maximizing their expected rewards. While this objective is effective in driving agents toward high-return strategies, it completely

neglects the notion of diversity. As a result, when the reward function implicitly conflicts with certain behaviors, agents may converge on a narrow set of solutions that exploit the reward signal while ignoring the broader behavioral space that does not maximize rewards [33]. To illustrate, consider the task of training robots to explore different positions within a house. From a QD perspective, we would like to discover a collection of agents, each specializing in reaching a distinct location, thereby achieving wide coverage of possible behaviors. However, if the reward function emphasizes energy preservation, encouraging agents to minimize movement, the optimal RL strategy quickly degenerates into agents remaining at their initial starting positions. From the standpoint of reward maximization, this is indeed the best outcome, but it leads to poor behavioral diversity of the environment.

To address this limitation, a recent method known as Descriptor-Conditioned RL MAP-Elites (DCRL-ME) has been proposed. The key idea is to enrich the learning process by encourging the agent's policy not only to optimize given rewards but also stays close to the desired type of behavior (e.g., a specific target location in the house). To achieve this, they condition a critic network (a network that estimates the total future rewards of an action) with the target descriptors and weight the immediate rewards with the closeness of current behavior to the this target descriptors. This simple yet powerful modification prevents the collapse of diversity, as each agent is encouraged to specialize in achieving high-quality solutions for its assigned behavioral descriptor. In this way, DCRL-ME effectively combines the strengths of reinforcement learning with the diversity-preserving principles of QD algorithms.

In the original formulation, the DCRL-ME framework employs Twin-Delayed Deep Deterministic Policy Gradient (TD3) [16] as its reinforcement learning backbone. While TD3 has been empirically shown to be effective across a variety of benchmarks, its limitations become apparent in high-dimensional control tasks with many local optima. In such settings, TD3 often struggles to scale effectively, a phenomenon that has been documented

not only in standalone RL studies [17, 18] but also in other QD methods that rely on TD3 as their learning component [23].

To address this issue, we propose substituting TD3 with the Soft Actor-Critic (SAC) [17, 18] algorithm, resulting in a new variant that we call DCSAC-ME. SAC offers several advantages over TD3, most notably its entropy-regularized objective, which encourages exploration and helps mitigate premature convergence to sub-optimal solutions. This makes SAC a promising candidate for high-dimensional environments where behavioral diversity and robustness are crucial. However, directly incorporating SAC into the DCRL-ME framework often fails to perform well due to SAC's sensitivity to reward scaling, a well-known challenge in continuous control tasks. To examine this problem, we also evaluate DCSAC-ME under a range of reward-scaling hyperparameters across three challenging environments and provide an intuitive way to select such hyperparameter values. Our results demonstrate that DCSAC-ME substantially outperforms the original TD3-based approach. This highlights the scalability and robustness of SAC when properly adapted to the QD setting, and establishes DCSAC-ME as a strong alternative for learning diverse, high-quality policies in complex domains.

Our main contributions are summarized as follows:

- **A novel integration of SAC into the DCRL-ME framework.** We replace TD3 with SAC to better handle high-dimensional continuous control tasks, benefiting from SAC's entropy-regularized objective that promotes exploration and robustness.
- **A careful treatment of SAC's sensitivity to reward scaling.** We show that a naive substitution of SAC into DCRL-ME performs poorly due to reward-scaling issues. To address this, we systematically evaluate DCSAC-ME under a range of scaling hyperparameters across multiple environments.
- **Substantial improvements over baseline.** Our experiments demonstrate that DCSAC-ME consistently outperforms DCTD3-ME,

achieving particularly strong gains in the Humanoid environment, where it delivers up to 8× higher performance. This result underscores the scalability of SAC in QD frameworks and its effectiveness in discovering diverse, high-quality solutions in high-dimensional domains.

II. RELATED WORKS

2.1. Deep Reinforcement Learning for Continuous Control

RL [27, 29] has become a powerful framework for solving continuous control problems, which are central to robotics, locomotion, and other domains where agents must operate in high-dimensional, continuous action spaces. These problems pose unique challenges, as algorithms must not only learn effective control strategies but also generate smooth and stable actions. A major step forward came with the introduction of the Deterministic Policy Gradient (DPG) [28] algorithm, which extended the policy gradient framework to deterministic policies. Building on this, Deep Deterministic Policy Gradient (DDPG) [20] combined deterministic policy gradients with deep neural networks and off-policy learning. DDPG demonstrated impressive performance on continuous control tasks but is known to be overestimated its action values. To address this DDPG's limitations, Twin Delayed Deep Deterministic Policy Gradient (TD3) [16] introduced clipped double Q-learning to reduce overestimation bias, and delayed policy updates with target policy smoothing to stabilize learning. These improvements made TD3 a stronger and more stable baseline for continuous control, especially in environments with sparse or noisy rewards. Parallel to these off-policy methods, Proximal Policy Optimization (PPO) [35] was developed to stabilize on-policy learning. By constraining or clipping policy updates, these algorithms achieved more reliable training and became widely used in both research and practical applications. However, they are often less sample-efficient than off-policy counterparts. Another influential contribution is Soft Actor-Critic (SAC) [17, 18], which incorporates the maximum entropy framework to encourage exploration by optimizing both expected reward and entropy. SAC benefits from the stability of stochastic policies, off-policy replay buffers, and sample efficiency, making it one of the most widely adopted algorithms for continuous control today.

2.2. Quality-Diversity

Quality-Diversity (QD) [3, 7, 25] algorithms are designed to produce collections of solutions that are both diverse and high-performing, rather than converging to a single optimum. This perspective has gained traction because diversity provides robustness, adaptability, and the potential for reusing learned behaviors in novel or changing environments. QD algorithms have been successfully applied to problems such as robot damage recovery [4, 6], where agents can adapt by switching to alternative behaviors when parts of the robot malfunction, and overcoming deceptive reward landscapes, where traditional RL or EA methods might converge to poor local optima. By fostering behavioral diversity, QD methods avoid the trap of over-specialization in reward function providing deceptive signals and provide a toolkit of possible solutions that can be deployed flexibly [34]. Recent studies highlight that Quality-Diversity (QD) methods offer a competitive alternative for enabling robots to acquire a wide archive of diverse skills [1]. Importantly, these skills are not isolated; they can later be composed in a hierarchical manner, allowing the robot to flexibly combine and adapt them in order to act effectively when faced with novel or unforeseen environments [1].

One of the foundational methods in this area is MAP-Elites [21], which discretizes a user-defined descriptor space into a grid and searches for the best-performing solution in each cell. MAP-Elites has been shown to be effective in robotics control [4, 11, 22, 33], damage recovery [4, 6]. However, its reliance on undirected genetic operators such as mutation and crossover can lead to slow convergence, especially in high-dimensional descriptor spaces. Several extensions have been introduced to address these limitations. CVT-MAP-Elites [32] improves scalability by replacing the grid with a centroidal Voronoi tessellation, allowing better coverage in high-dimensional spaces. Other approaches incorporate surrogate models to reduce evaluation costs or gradient-based search mechanisms to accelerate convergence and improve performance [5, 13, 14, 30, 31].

A more recent line of work integrates QD with RL [12, 23, 24]. Algorithms such as Policy Gradient Assisted MAP-Elites (PGA-MAP-Elites) [22] and Descriptor-Conditioned Reinforcement Learning MAP-Elites (DCRL-ME) [33] leverage policy gradient updates in place of random mutations, resulting in significantly improved efficiency in continuous control tasks. These approaches and others demonstrate the potential of combining directed optimization from RL with the exploratory power of QD, leading to archives of behaviors that are both diverse and high-performing.

In this report, we further extend the DCRL-ME framework by substituting the original TD3 algorithm with a more robust and explorative SAC method. This allows the framework to learn effectively good agents on high-dimensional environments with many local optimums.

III. BACKGROUND

3.1. Problem Formulation

Here, we follow standard RL formulation by adopting Markov Decision Process (MDP), where an agent interacts with an environment at discrete time steps $\boldsymbol{t}$. At each step $\boldsymbol{t}$, the agent observes a state $\boldsymbol{s_t} \in \boldsymbol{S}$, selects an action $\boldsymbol{a_t} \in \boldsymbol{A}$, and receives a scalar reward $\boldsymbol{r_t} = \boldsymbol{r}(\boldsymbol{s_t}, \boldsymbol{a_t}) \in \boldsymbol{R}$. The environment then transitions to the next state $\boldsymbol{s_{t+1}} \in \boldsymbol{S}$ according to the transition dynamics $\boldsymbol{s_{t+1}} \sim \boldsymbol{p}(. |\boldsymbol{s_t}, \boldsymbol{a_t})$. Repeated interactions generate a trajectory of states, actions, and rewards. In this work, we focus on both two settings for the policy, deterministic policies (output only one action) and stochastic policies (output a distribution of actions to sample from). Most of prior studies in QD work with deterministic policies to consistently leverage deterministic policy gradient algorithm like TD3 [16]. However, we consider stochastic policies as well to leverage SAC, which operates based on stochastic output actions. A policy parameterized by $\boldsymbol{\phi}$, denoted $\boldsymbol{\pi_\phi} \in \boldsymbol{\Pi}: \boldsymbol{S} \rightarrow \boldsymbol{A}$, mapping from states to actions. Normally, a policy is a neural network (MLP in this case) to enable learning in high-dimensional space.

Within this RL setting, **Quality-Diversity** (QD) [25] algorithms aim to discover a collection of diverse yet high-performing policies. The fitness function $\boldsymbol{F}: \boldsymbol{\Pi} \rightarrow \mathbf{R}$ measures the performance of a policy and is typically defined as the expected cumulative reward over an episode of length T:

$$\boldsymbol{F}(\boldsymbol{\pi}) = \boldsymbol{E}_{\boldsymbol{\pi}}[\boldsymbol{\Sigma}_{\boldsymbol{t=0}}^{\boldsymbol{T-1}} \boldsymbol{r_t}]$$

To capture behavioral diversity, a *descriptor function* $\boldsymbol{D}: \boldsymbol{\Pi} \rightarrow \boldsymbol{D} \subseteq \boldsymbol{R^d}$ maps each policy into a user-defined descriptor space $\boldsymbol{D}$, which encodes behavioral $\boldsymbol{d} \in \boldsymbol{N}$ characteristics of solutions relevant to the task. The descriptor space can be intuitively understood as some features that we can extract from the agents (e.g., the xy coordination or the length of the legs of a robot). The overall objective of QD algorithms in this context is to identify, for each region of the descriptor space, the highest-fitness policy available.

Equivalently, the goal is to evolve a population of solutions that are both high-performing with respect to $\boldsymbol{F}$ and diverse with respect to $\boldsymbol{D}$.

3.2. MAP-Elites

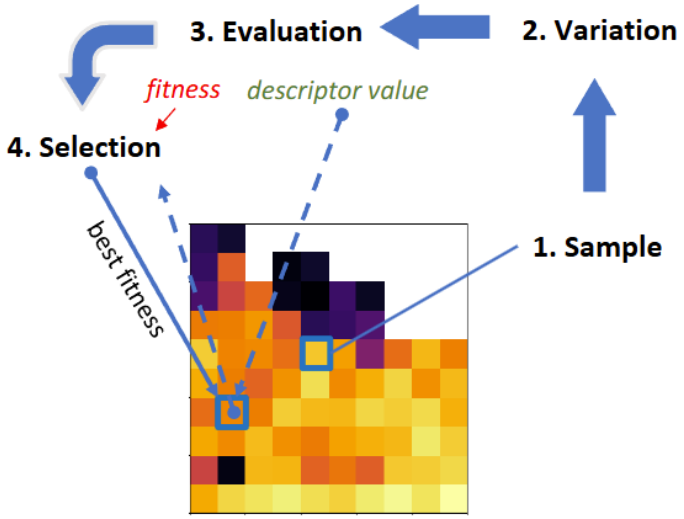


Figure 2: *An iteration of MAP-Elites algorithm [21].*

MAP-Elites [21] is a foundational QD algorithm designed to explore a descriptor space $\boldsymbol{D}$ by maintaining an archive $\boldsymbol{X}$, which is a multi-dimensional grid of cells. Each cell represents a niche in the behavioral space, and the objective is to fill the archive with high-quality solutions that cover as many niches as possible. The algorithm begins with an empty archive and a set of randomly initialized solutions. At each iteration, MAP-Elites repeats a simple evolutionary loop (illustrated in Figure 2):

- **Sample:** a batch of solutions is uniformly sampled from the archive.
- **Variation:** the selected solutions are mutated or crossovered to generate new offspring.
- **Evaluation and Selection:** each offspring is evaluated for both its fitness and its descriptors. Based on the descriptors, the offspring is mapped to a specific cell in the archive. If that cell is empty, the offspring is inserted. If it is already occupied, the offspring replaces the currently stored elite only if it has higher fitness.

Over time, this process populates the archive with diverse and high-performing solutions across the descriptor space, resulting in a archive of specialized behaviors rather than a single optimized policy. Despite its simplicity and effectiveness, MAP-Elites has a key limitation. The search process is driven by undirected variation operators (e.g., random mutation, crossover), which are not informed by the fitness gradient. This makes the search highly divergent but inefficient in high-dimensional spaces.

3.3. Descriptor-Conditioned RL MAP Elites

Descriptor-Conditioned RL MAP-Elites (DCRL-MAP-Elites) [33] extends MAP-Elites by leveraging policy gradient formulation in RL as a directed variation operator. Instead of relying solely on random mutation and crossover, DCRL-MAP-Elites integrates a descriptor-conditioned actor-critic framework, which enables faster convergence and smoother generalization across descriptors. The DCRL-ME operation process is provided in Algorithm 1 (originated from the [33]).

Descriptor-Conditioned Critic. To effectively optimize the policy using RL framework, we need a way to determine the total future rewards of the action given current state (called action-value). Here, we use a critic, which is a neural network (or any arbitrary function) to approximate such value. Besides taking the state and action as input, the critic is extended to condition on a target descriptor $\boldsymbol{d} \in \boldsymbol{D}$, leading to the descriptor-conditioned value function $\boldsymbol{Q_\theta(s, a|d)}$. Given a trajectory with observed descriptor $\boldsymbol{d}$, the critic update incorporates a similarity score $\boldsymbol{S(d, d')}$ between the *target* descriptor $\boldsymbol{d}$ and the actual *observed* descriptor $\boldsymbol{d'}$. Formally, the critic is trained on transitions $(\boldsymbol{s_t, a_t, r_t, s_{t+1}, d'})$ sampled from a replay buffer $\boldsymbol{B}$ through the L2 loss:

$$\boldsymbol{L(\theta_i) = E_{(s_t, a_t, r_t, s_{t+1}, d') \sim B}[\left(Q_{\theta_i}(s_t, a_t, d) - y_t\right)^2]},$$

with the target:

$$\mathbf{y_t} = \mathbf{S(d, d')r_t} + \boldsymbol{\gamma}[\min_{j=\{1,2\}} \mathbf{Q}_{\bar{\theta}_j}(\mathbf{s}_{t+1}, \boldsymbol{\pi}(\mathbf{s_t}, \boldsymbol{d}), \boldsymbol{d})]$$

where $\bar{\boldsymbol{\theta}}_{\boldsymbol{j}}$ denotes the target critic parameters, which are updated towards the original critic through exponetial moving average. The minimum operator in this equation is originated from TD3 as a solution to overestimation bias problem. Here, the similarity score $\mathbf{S(d, d')}$ encourages the critic to assign larger values to actions that are directed towards the target descriptors by weighting returns based on descriptor closeness.

Archive Distilation via Descriptor-Conditioned Actor. Besides the policies stored in the archive, DCRL-ME also has another external policy (which is not included in the archive) used to trained alongside with the critic and distil all behaviors in the archive with only one network. The actor is defined as a descriptor-conditioned policy $\boldsymbol{\pi_\phi(s, d)}$, which maps both the current state $\boldsymbol{s}$ and a target descriptor $\boldsymbol{d}$ to action. It is optimized to maximize the descriptor-conditioned value function:

$$\mathbf{J_\pi(\phi)} = \mathbf{E}_{s_t, d\sim B, a_t\sim\pi_\phi,}[\mathbf{Q}_{\theta_i}(\mathbf{s_t}, \mathbf{a_t}, \boldsymbol{d})],$$

This formulation allows the external actor to represent a diverse set of behaviors within a single neural policy by conditioning on $\boldsymbol{d}$.

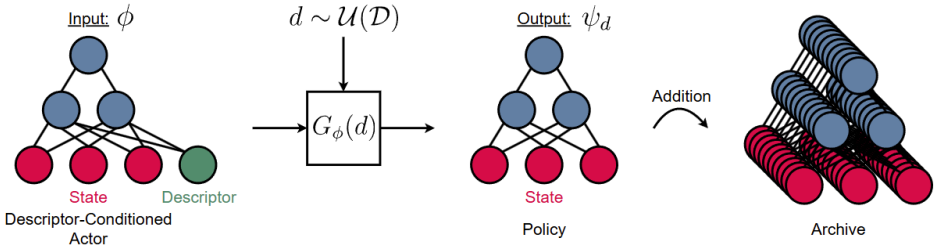


Figure 3: *Illustration of functional process for Actor Injection mechanism. Image originates from [33]*

Actor Injection. Besides the genetic and policy-gradient-based mutation, DCRL-MAP-Elites introduces an *actor injection* (Figure 3) to effectively generate new offspring. For a given target descriptor $\boldsymbol{d}$, the descriptor-conditioned actor $\boldsymbol{\pi_\phi(s, d)}$ is modified into a deterministic, unconditioned policy $\boldsymbol{\pi(s)}$, which is then evaluated in the environment and inserted into the

archive if it improves upon the incumbent solution in the corresponding cell. This ensures that the archive consists of stand-alone policies, while the descriptor-conditioned actor serves as a compact generator of candidate solutions.

Algorithm 1 DCRL-MAP-Elites

Require: GA batch size b_{GA}, PG batch size b_{PG}, Actor Injection batch size b_{AI}, total batch size $b = b_{\text{GA}} + b_{\text{PG}} + b_{\text{AI}}$
Initialize archive $\mathcal{X}$ with b random solutions and replay buffer $\mathcal{B}$
Initialize critic networks Q_{θ_1}, Q_{θ_2} and actor network π_ϕ
$i \leftarrow 0$
while $i < I$ **do**
  TRAIN_ACTOR_CRITIC$(\pi_\phi, Q_{\theta_1}, Q_{\theta_2}, \mathcal{B})$
  $\pi_{\psi_1}, \dots, \pi_{\psi_b} \leftarrow$ SELECTION$(\mathcal{X})$
  $\pi_{\widehat{\psi}_1}, \dots, \pi_{\widehat{\psi}_{b_{\text{GA}}}} \leftarrow$ VARIATION_GA$(\pi_{\psi_1}, \dots, \pi_{\psi_{b_{\text{GA}}}})$
  $\pi_{\widehat{\psi}_{b_{\text{GA}}+1}}, \dots, \pi_{\widehat{\psi}_{b_{\text{GA}}+b_{\text{PG}}}} \leftarrow$ VARIATION_PG$(\pi_{\psi_{b_{\text{GA}}+1}}, \dots, \pi_{\psi_{b_{\text{GA}}+b_{\text{PG}}}}, Q_{\theta_1}, \mathcal{B})$
  $\pi_{\widehat{\psi}_{b_{\text{GA}}+b_{\text{PG}}+1}}, \dots, \pi_{\widehat{\psi}_b} \leftarrow$ ACTOR_INJECTION(π_ϕ)
  ADDITION$(\pi_{\widehat{\psi}_1}, \dots, \pi_{\widehat{\psi}_b}, \mathcal{X}, \mathcal{B})$
  $i \leftarrow i + b$
function ADDITION$(\pi_{\widehat{\psi}} \dots, \mathcal{X}, \mathcal{B})$
  for $\pi_{\widehat{\psi}} \dots$ **do**
    $(f, \text{transitions}) \leftarrow F(\pi_{\widehat{\psi}}), d \leftarrow D(\pi_{\widehat{\psi}})$
    INSERT$(\mathcal{B}, \text{transitions})$
    if $\mathcal{X}(d) = \emptyset$ or $F(\mathcal{X}(d)) < f$ **then**
      $\mathcal{X}(d) \leftarrow \pi_{\widehat{\psi}}$

3.4. Soft Actor Critic

Soft Actor-Critic (SAC) [17, 18] is an off-policy actor-critic algorithm that has become one of the most effective methods for continuous control. Unlike traditional reinforcement learning algorithms that only maximize the expected return, SAC follows the principle of *maximum entropy* RL. In this setting, the agent aims to maximize both the cumulative rewards and the entropy of its policy (i.e., the randomness of selected action), which encourages persistent exploration and prevents premature convergence to suboptimal behaviors.

IV. PROPOSED METHOD

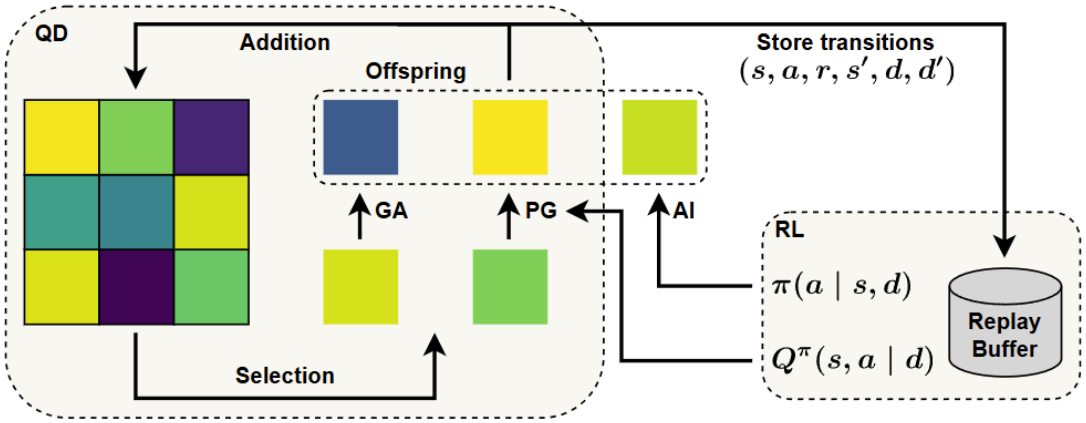


Figure 4: *An illustration of how DCRL-ME works. The image originate from [33]*

In this section, we introduce our proposed method, which integrates the DCRL-ME framework with Soft Actor-Critic (SAC) as the RL backbone. The objective of this design is to leverage SAC's entropy-regularized optimization to produce agents that not only achieve high performance but also maintain behavioral diversity through descriptor conditioning. By conditioning both the actor and the critic on target descriptors, the algorithm encourages agents to specialize in distinct behavioral niches, while SAC's stochastic policy and stable critic updates provide robustness and scalability in high-dimensional continuous control tasks.

The overall operation of our method follows the same pipeline as the original DCRL-ME framework. As illustrated in Figure 4, the process begins with the selection of parent policies from the archive, which are then mutated through a combination of genetic variation, RL-based updates and actor injection mechanism to generate a batch of new offspring. These offsprings are then evaluated by running in the environment in T timesteps and return their fitness and descriptor values. Moreover, all experiences collected through interation are stored in a replay buffer for later training. The descriptor-conditioned critics guide the optimization by weighing rewards according to descriptor similarity, ensuring that new offspring remain consistent with their assigned niches. This integration of SAC into the DCRL-

ME framework forms the basis of our proposed approach, which we refer to as DCRL-ME (SAC).

4.1. Stochastic Policies

We first reformulate the policy to be stochastic for both the external descriptor-conditioned policy $\boldsymbol{\pi_\phi(a|s,d)}$ and all the unconditioned policies within the archive $\boldsymbol{\pi_\phi(a|s)}$. Specifically, the policy takes as input the current state (with the desired descriptor), and outputs a distribution over actions, instead of fixed, deterministic actions. We assume this distribution to be Gaussian, such that the policy only needs to output the mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$, which together define a normal distribution. To generate actions, we first construct the Gaussian distribution parameterized by $(\boldsymbol{\mu, \sigma})$, sample from it, and then apply a tanh transformation to ensure that the resulting actions lie within the bounded action space [-1; 1]. However, directly sampling from (μ, σ) would prevent gradients from propagating through the sampling step. To overcome this, we employ the reparameterization trick, where the action is computed as $\boldsymbol{a = \mu + \sigma\epsilon, \epsilon \sim N(0; I)}$. This formulation enables gradients to flow back through the policy network parameters during optimization, making the stochastic descriptor-conditioned policy both expressive and trainable.

4.2. Training descriptor-conditioned actor-critic with SAC

To optimize the descriptor-conditioned and unconditioned policies, we maintains two soft Q-functions $\boldsymbol{Q_{\theta_1}}$ and $\boldsymbol{Q_{\theta_2}}$ like in SAC. The descriptor-conditioned critics are trained to minimize the soft Bellman error:

$$\boldsymbol{L(\theta_i) = E_{(s_t,a_t,r_t,s_{t+1})\sim B}[(Q_i(s_t,a_t,d) - y_t)^2]},$$

where the target is defined as:

$$\mathbf{y_t = S(d,d')r_t + \gamma[\min_{j=\{1,2\}} Q_{\bar{\theta}_j}(s_{t+1},a_{t+1},d) - \alpha \log \pi_\phi(a_{t+1}|s_{t+1},d)]}$$

with $\boldsymbol{a_{t+1} \sim \pi_\phi(\cdot\,|s_{t+1},d)}$ and $\boldsymbol{\bar{\theta}_j}$ denoting the target critic parameters. This formulation integrates both the estimated value and the entropy regularization directly into the critic update. Moreover, it weight the immediate reward with

the similarity between the target descriptors $\boldsymbol{d}$ and the observed descriptors $\boldsymbol{d'}$, forcing the agents to remain near the desired behaviors. The similarity is is modeled using the kernel function $\boldsymbol{S(d, d')} = \mathbf{exp}\ (\frac{-\left|\left|\boldsymbol{d}-\boldsymbol{d'}\right|\right|_{\boldsymbol{D}}}{\boldsymbol{L}})$, where $\boldsymbol{L}$ is the length scale hyperparameter controlling the smoothness of the similarity. A smaller value of $\boldsymbol{L}$ enforces stricter adherence to the target descriptor, as even small deviations between $\boldsymbol{d}$ and $\boldsymbol{d'}$ sharply reduce the weight of the reward. Conversely, a larger value of $\boldsymbol{L}$ allows broader generalization by smoothing the similarity landscape, enabling the critic to propagate reward signals across a wider region of the descriptor space. The descriptor-conditioned actor and parent policies selected from the archive is then updated by minimizing:

$$\mathbf{J_{\pi}(\phi)} = \mathbf{E}_{s_t,d\sim B,a_t\sim\pi_{\phi}}[\boldsymbol{\alpha}\mathbf{log}\,\boldsymbol{\pi}_{\boldsymbol{\phi}}(\mathbf{a_t}|\mathbf{s_t},\mathbf{d}) - \min_{i=\{1.2\}} \mathbf{Q}_{\boldsymbol{\theta}_i}(\mathbf{s_t},\mathbf{a_t},\boldsymbol{d})], \tag{1}$$

where $\boldsymbol{H}\big(\boldsymbol{\pi}(\cdot\,|\boldsymbol{s_t})\big) = -\boldsymbol{E}_{\boldsymbol{a_t}\sim\boldsymbol{\pi}}[\boldsymbol{log\pi}(\boldsymbol{a_t}|\boldsymbol{s_t})]$ denotes the policy entropy and $\boldsymbol{\alpha}$ is a temperature parameter that balances reward maximization against entropy maximization. This objective encourages the policy to favor actions that not only yield high Q-values but also maintain adequate entropy and stay close to the target descriptors.

An important component of SAC is the temperature parameter $\boldsymbol{\alpha}$, which controls the trade-off between reward maximization and exploration. Instead of fixing this parameter manually, we can learn it automatically. We also follow this procedure and corperate this automatic mechanism into our modified DCSAC-ME algorithm by minimizing the following objective:

$$\mathbf{J}(\boldsymbol{\alpha}) = \mathbf{E}_{\boldsymbol{a_t}\sim\boldsymbol{\pi_{\phi}},\boldsymbol{s_t},\boldsymbol{d'}\sim\boldsymbol{B}}[-\boldsymbol{\alpha}(\mathbf{log}\boldsymbol{\pi}_{\boldsymbol{\phi}}(\mathbf{a_t}|\mathbf{s_t},\mathbf{d'}) + \mathbf{H}_{\boldsymbol{target}}],$$

where $\mathbf{H}_{\boldsymbol{target}}$ is a predefined target entropy. This formulation encourages the policy entropy to match the desired level of randomness, allowing the agent to adjust its exploration adaptively during training, eliminate the burden of manually tuning a sensitive hyperparameter $\boldsymbol{\alpha}$.

4.3. Variation operators

At each generation, we generate a batch of new offprings from the parent networks sampled from the archive. Following the original framework, we apply three distinct operators to introduce variation into the parents.

The first operator is a genetic algorithm (GA)–based mutation known as Isoline-DD, a directional mutation mechanism that perturbs the parameters of the parent policy along isotropic and data-driven directions. This operator introduces exploratory diversity while maintaining a degree of structure, enabling the offspring to explore new regions of the parameter space without deviating too far from their parent solutions:

$$\widehat{\boldsymbol{\phi}} = \boldsymbol{\phi_1} + \boldsymbol{\sigma_1} \boldsymbol{N}(\mathbf{0}; \boldsymbol{I}) + \boldsymbol{\sigma_2}(\boldsymbol{\phi_2} - \boldsymbol{\phi_1}) \boldsymbol{N}(\mathbf{0}; \boldsymbol{I})$$

where $\boldsymbol{\phi_1}$ and $\boldsymbol{\phi_2}$ are two parent networks parameters randomly sampled from the archive, $\boldsymbol{\sigma_1}$ and $\boldsymbol{\sigma_2}$ are hyperparameter values managing the scale of randomness in mutation.

The second operators is policy gradient (PG) where we update the sampled parents from the archive with the following equation:

$$\mathbf{J}_{\boldsymbol{\pi}}(\boldsymbol{\phi}) = \mathbf{E}_{\boldsymbol{s_t} \sim \boldsymbol{B}, \boldsymbol{a_t} \sim \boldsymbol{\pi_\phi}} [\boldsymbol{\alpha} \mathbf{log}\, \boldsymbol{\pi}_{\boldsymbol{\phi}}(\mathbf{a_t}|\mathbf{s_t}) - \min_{\boldsymbol{i}=\{\mathbf{1.2}\}} \mathbf{Q}_{\boldsymbol{\theta_i}}(\mathbf{s_t}, \mathbf{a_t}, \boldsymbol{d_p})],$$

Note that this equation is different from Equation (1) in two aspects. The policy $\boldsymbol{\pi_\theta}(\boldsymbol{a_t}|\boldsymbol{s_t})$ is not the external actor but the parent policy from the archive. Additionally, the target descriptors is not sampled from the replay buffer but from the previous descriptors of the parents. This PG mutation enables the quality of the agents to be boosted up rapidly while still maintaining at the current position in the descriptor space.

The third variation operator we utilize is the actor injection which inject the parameters of the external descriptor-conditioned actor network to some of the parent policy networks to directly enhance the their performance without further training or modification.

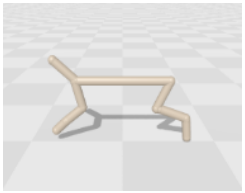
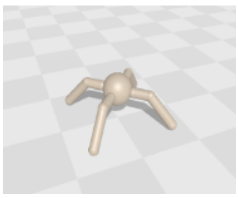
V. EXPERIMENTS

5.1. Experiment Setups

5.1.1. Benchmarks

We conduct evaluations across three continuous control tasks drawn from the Brax [15] physics simulator: HalfCheetah-Uni, Ant-Omni, and Humanoid-Uni. The numerical specifications of these environments, including their state/action and descriptor dimensions, are summarized in Table 1. These environments are chosen as our primary benchmarks because they differ significantly in complexity, reward structures, and behavioral descriptors, thereby providing a comprehensive testbed for our proposed method along with other baselines.

Table 1: *Information about state, action, descriptor dimensions, and episode length and illustration examples of three locomotion tasks: Halfcheetah-Uni, Ant-Omni, and Humanoid-Uni*

	HalfCheetah-Uni	**Ant-Omni**	**Humanoid-Uni**
			
State dim	19	30	245
Action dim	6	8	17
Descriptor dim		2	
Episode len	1000	250	1000

The "-Uni" tasks (unidirectional) require agents to move forward along the x-axis as quickly as possible. For these tasks, we define behavioral descriptors based on the ground contact ratio of the agents' leg tips; specifically, the two legs of the HalfCheetah and the two feet of the Humanoid. These descriptors capture meaningful variations in locomotion strategies while still allowing for reward optimization toward forward

velocity. The "-Omni" tasks (omnidirectional), in contrast, allow agents to move freely in the two-dimensional xy-plane. Here, the objective is to optimize for energy preservation, i.e., minimizing unnecessary movement. The corresponding behavioral descriptors are defined as the agent's final position on the xy-plane relative to its initial spawn location, effectively capturing a broad spectrum of locomotion behaviors that may emerge while satisfying the energy-preservation reward. This environment contains a conflict between the fitness (quality) and a large portion of behaviors which may hinder the diversity if the mutation only cares about quality.

5.1.2. Metrics

For quantitative evaluation, we adopt three widely used metrics in the Quality-Diversity (QD) literature: Coverage, Max Fitness, and QD Score.

- **Coverage** measures the fraction of niches in the archive that are occupied by at least one solution. This metric emphasizes the diversity of behaviors, as higher coverage indicates a broader exploration of the behavioral space.
- **Max Fitness** captures the quality dimension by reporting the maximum fitness (i.e., cumulative reward) achieved by the best individual across the entire archive. This provides a direct indicator of the peak performance attainable by the method.
- **QD Score** combines both quality and diversity into a single metric. It is computed as the sum of fitness values across all individuals in the archive, assigning a score of zero to empty niches. As a result, QD Score rewards algorithms that not only discover diverse solutions but also ensure that those solutions are of high quality. Since this formulation requires fitness values to be non-negative, we follow prior work by applying a reward transformation: at each step, the raw rewards are offset by the worst possible reward so that the resulting values are always greater than or equal to zero. This ensures that QD Score is computed consistently across all tasks.

5.1.3. Baselines and Hyperparameters

To evaluate the effectiveness of our proposed method, we conduct a comparative study against three representative baselines:

- **MAP-Elites [21],** the canonical QD algorithm that operates without any RL component. In our experiments, we implement MAP-Elites with the Isoline-DD mutation operator which perturbs existing solutions to generate offspring. This baseline provides a test of whether a purely evolutionary approach, relying only on undirected genetic variation, is sufficient to generate both diversity and quality in the evaluated environments.
- **Policy Gradient Assisted MAP-Elites (PGA-ME) [22]** which augments MAP-Elites with RL as a directed mutation mechanism. Specifically, it employs TD3 as the RL backbone to locally improve selected parent solutions. By combining the exploratory power of MAP-Elites with the exploitative optimization of TD3, PGA-ME represents a more sample-efficient alternative to pure evolutionary search. This baseline allows us to test the extent to which incorporating RL updates can accelerate convergence and improve the overall quality of solutions.
- **DCRL-ME (TD3) [33]** which extends the integration of RL into QD one step further. Unlike PGA-ME, DCRL-ME explicitly conditions both the actor and critic on a target descriptor, and modifies the optimization objective to include a descriptor similarity function. This design encourages policies not only to maximize rewards but also to remain consistent with their assigned descriptors, thereby ensuring that the archive is populated with solutions that are both high-performing and diverse across behavioral niches.

To implement all methods, we use the QDax [2] library, which is implemented in JAX to accelerate the search process by vectorizing functions. All methods used CVT-MAP-Elites [32] to initialize the structure of the archive. The policy (both descriptor-conditioned and unconditioned versions)

is an MLP network with two hidden layers sized (128, 128). The critic is also an MLP with two hidden layers sized (256, 256). At each iteration, we randomly sample a batch of 256 parent policies from the archive and perform variation. For MAP-Elites, all of them are pass through the GA mutation to create new offsprings. For PGA-MAP-Elites, half of the parents are mutated by GA mutation and the other half are optimized using policy gradient from the RL-based mutation. For DCRL-ME (both TD3 and SAC variants), 128 offsprings created by GA mutation, 64 created by policy-gradient mutation, and the rest created by actor injection mechanism. Other hyperparameters are used the same as in the original publication [33]. Each run has the budget of 1,000,000 evaluations.

5.2. Experimental Results

5.2.1. Quantitative Results

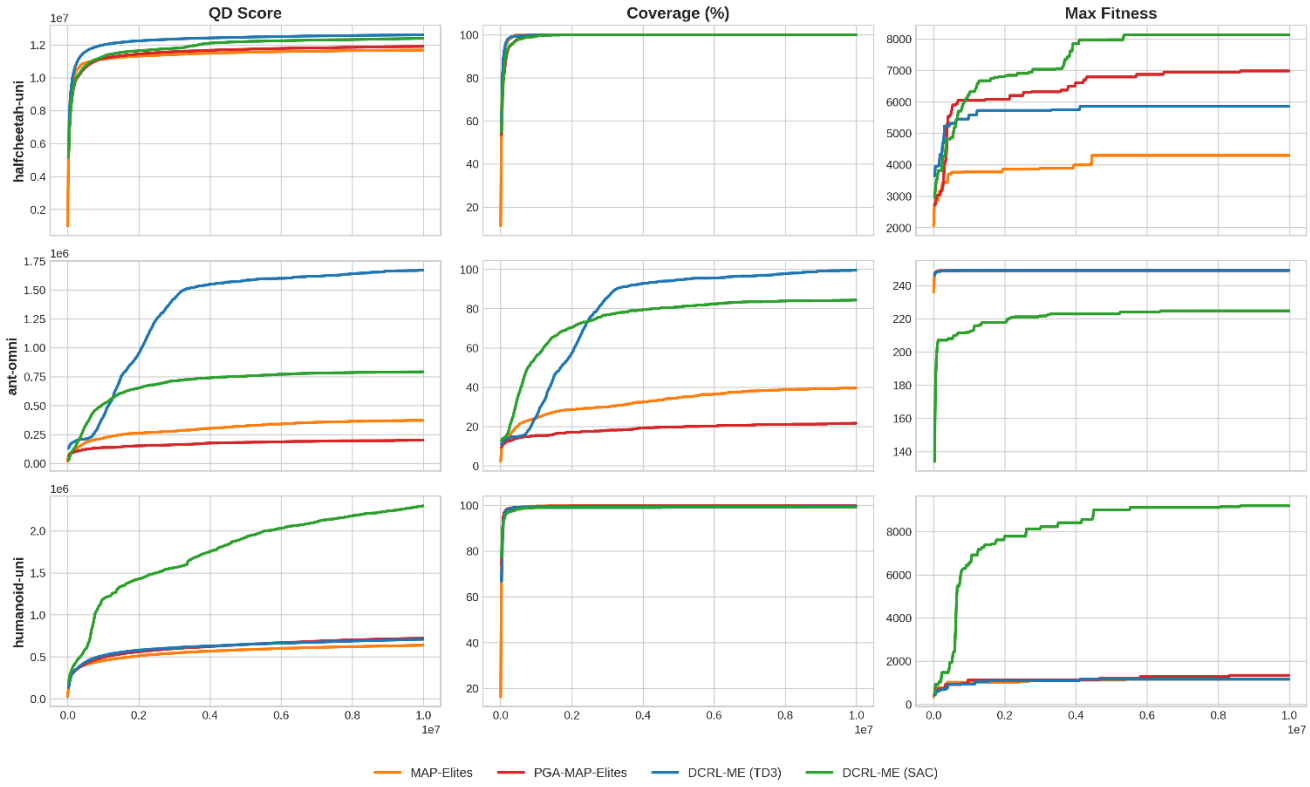


Figure 5: *Learning curves of QD Score, Coverage and Max fitness score across number of evaluations from four methods on three environments.*

We evaluate the proposed DCRL-ME (SAC) against baselines on three locomotion tasks. Performance is reported in Figure 5.

On Halfcheetah-Uni, a relatively low-dimensional task, all methods achieve near-complete coverage, suggesting that exploration is not the primary bottleneck. However, substantial differences emerge in terms of solution quality. Both variants of DCRL-ME significantly outperform MAP-Elites and PGA-ME, with the SAC-based approach yielding the highest Max Fitness and competitive QD Score. These results indicate that RL as a variation operator is critical for optimizing quality, and SAC provides more stable policy updates than TD3, resulting in higher overall performance.

Ant-Omni poses a greater challenge due to its conflict in quality and diversity. Here, MAP-Elites maintains moderate coverage, while PGA-ME struggles to improve the coverage due to its nature of enhacing only reward. DCRL-ME (TD3) achieves the highest coverage, nearly saturating the archive. By contrast, DCRL-ME (SAC) attains slightly lower coverage and substantially lower QD Score compared to its TD3-based counterpart. We hypothesize that this behavior arises from the use of stochastic policies and the entropy-regularized exploration mechanism in SAC. Instead of stopping once the agent reaches its target descriptor, the policy tends to continue random motions in the vicinity of the target. While this behavior preserves exploration, it also leads to unnecessary energy consumption, which in turn reduces the overall fitness of the solutions and consequently lowers the QD Score. We will consider fixing this behaviour as our future work.

The Humanoid-Uni task, characterized by high-dimensional state and action spaces and many local optima, provides the most stringent test of scalability. Both MAP-Elites and PGA-ME stagnate quickly, achieving low scores across all metrics. DCRL-ME (TD3) demonstrates moderate improvements but fails to substantially scale, with coverage plateauing early and fitness improvements stagnating. In contrast, DCRL-ME (SAC) achieves dramatic gains, reaching nearly complete coverage and substantially higher Max Fitness than all baselines. Notably, its QD Score surpasses DCRL-ME

(TD3) by approximately a **factor of eight**, showing the effectiveness of entropy-regularized exploration in avoiding premature convergence and scaling to high-dimensional continuous control.

Taken together, these results yield three main insights. First, undirected evolutionary search (MAP-Elites) achieves coverage but lacks the capacity to optimize quality effectively. Second, incorporating RL as a directed variation operator (PGA-ME, DCRL-ME) substantially improves quality, but the choice of RL backbone critically determines scalability. Finally, entropy-regularized learning with SAC not only stabilizes training and improves solution quality but also sustains exploration in high-dimensional environments where TD3-based methods collapse.

5.2.2. Qualitative Results

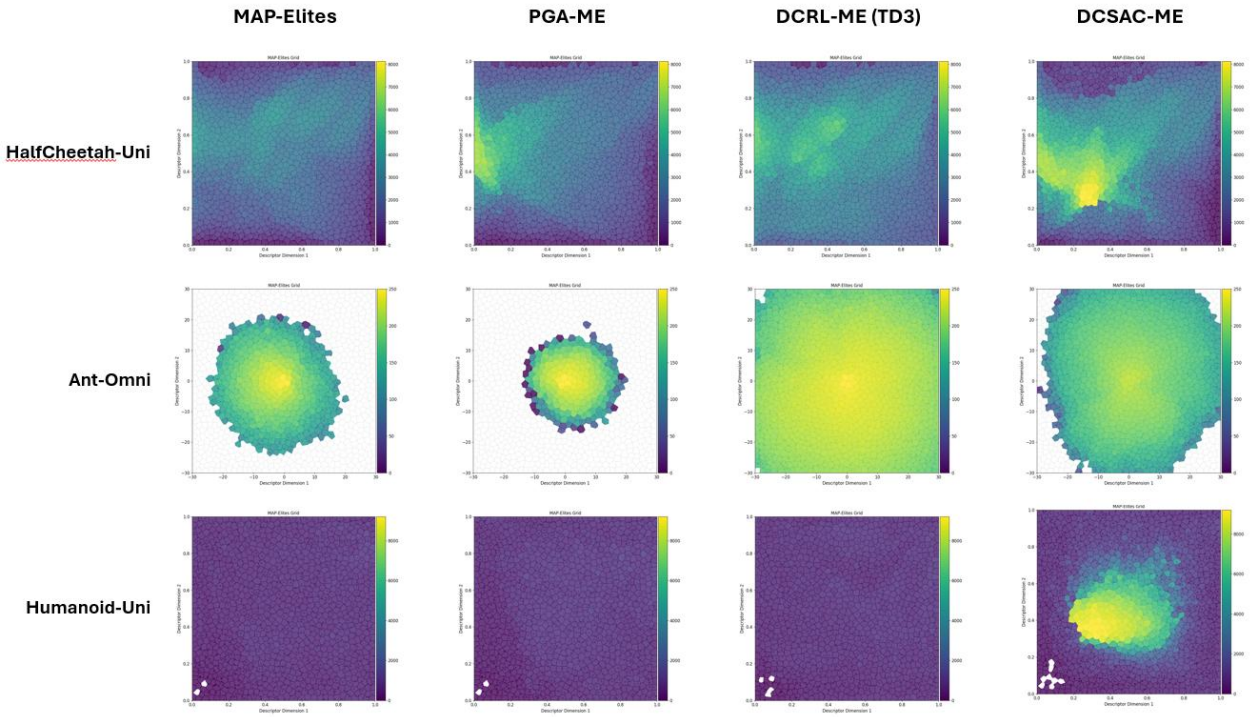


Figure 6: *Illustration of final returned archives from four methods across different tasks.* Each cell in the archive represents a distinct behavioral niche, with occupied niches shown as colored cells, where the intensity of the color reflects the quality (fitness) of the best solution found.

Archive comparison between methods.

Figure 6 visualizes the archives generated by MAP-Elites, PGA-ME, DCRL-ME (TD3), and our proposed DCSAC-ME across three tasks. Each cell in the archive corresponds to a behavioral niche, where non-white cells indicate occupancy and the brightness of the color encodes solution quality (fitness). Thus, a well-performing algorithm is expected to produce archives that are both densely populated (large coverage) and bright (high quality).

MAP-Elites produces archives with relatively broad coverage, as indicated by the large number of non-white cells across all environments. This is consistent with its divergent evolutionary search, which emphasizes exploration without strong guidance from the fitness signal. However, most cells remain dark-colored, particularly in Humanoid-Uni, reflecting low fitness solutions. This highlights the inefficiency of undirected genetic operators in improving quality in high-dimensional domains.

PGA-ME shows a different pattern where the occupied niches are generally brighter than those of MAP-Elites, confirming the role of policy gradient updates in pushing solutions toward higher returns. Nevertheless, the total number of non-white cells is lower, even compared to traditional MAP-Elites algorithm, especially in Ant-Omni. This reduction in coverage indicates that the algorithm tends to concentrate on reward-dense regions of the descriptor space, sacrificing exploration for exploitation in quality-diversity conflict environment like Ant-Omni.

The original DCRL-ME algorithm (with TD3 as RL backbone) attempts to balance these two aspects by conditioning policies and critics on descriptors. As a result, its archives contain more structured occupancy than PGA-ME, with specialized solutions in different descriptor regions. Yet, the archives reveal clear weaknesses in high-dimensional settings: in Humanoid-Uni, while the archive are dense, all occupied cells are still dark, showing that TD3 struggles to scale effectively. The limited exploration combined with premature convergence to sub-optimal solutions reduces both coverage and quality in these complex environments.

DCSAC-ME, by contrast, achieves the best balance. Its archives are densely filled with bright non-white cells, showing both broad coverage and high fitness across descriptor regions. The effect is particularly pronounced in Humanoid-Uni, where DCSAC-ME fills niches that remain empty under TD3-based methods and assigns them significantly higher quality. This demonstrates that SAC's entropy-regularized updates prevent premature convergence while enabling robust optimization of policy performance. The archives provide clear visual evidence that DCSAC-ME not only explores widely but also sustains high-quality solutions across the descriptor space.

Taken together, the archive comparisons confirm that entropy-regularized RL provides advantages for QD in high-dimensional continuous control. Whereas MAP-Elites prioritizes exploration, and PGA-ME/DCRL-ME struggle to scale in quality, DCSAC-ME consistently achieves both goals with high-quality and diverse solutions.

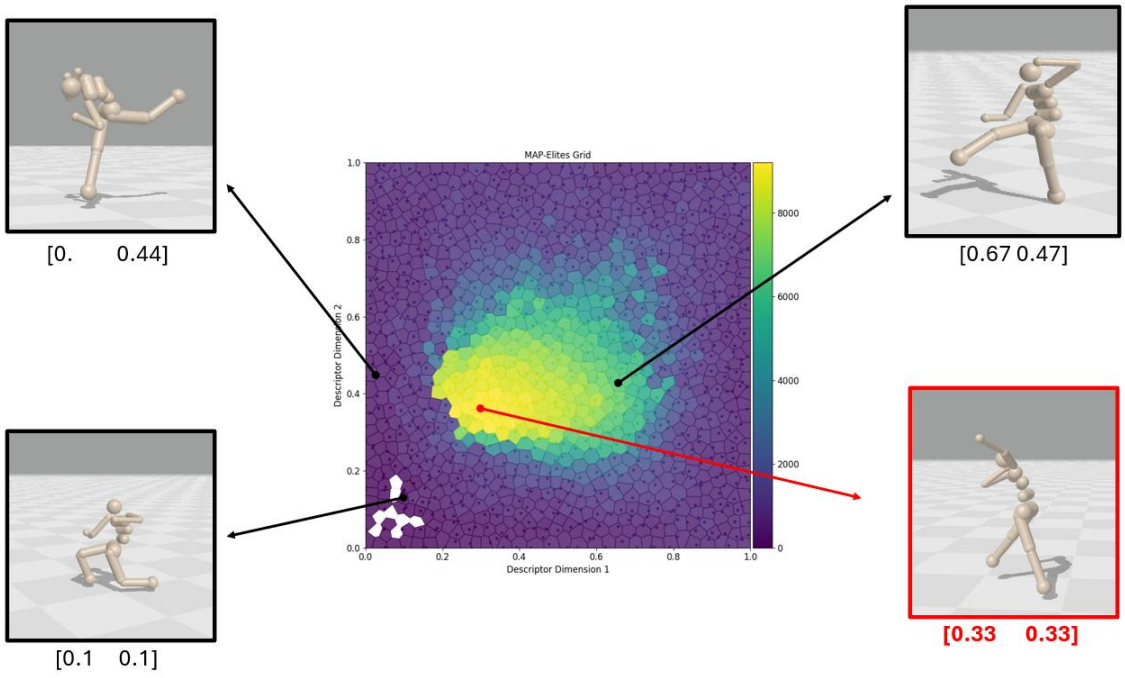


Figure 7: *Qualitative examples of found behaviors by DCRL-ME on Humanoid-Uni tasks, corresponding with their descriptor values shown below. The example with red frame indicates the highest-fitness agent in the archive.*

Examples of diverse behaviors of DCSAC-ME in Humanoid-Uni.

Figure 7 visualizes several representative behaviors extracted from the repertoire in the Humanoid-Uni task, where each agent corresponds to a distinct niche in the descriptor space. The examples highlight the breadth of behavioral diversity discovered by the algorithm. For instance, agents located near the bottom-left of the grid (e.g., descriptor [0.1, 0.1]) exhibit crouching or low-stance postures, while those in the upper region (e.g., [0.0, 0.44]) demonstrate upright balancing with extended legs. Moving toward the right-hand side of the repertoire, agents adopt leaning or stretched-out poses (e.g., [0.67, 0.47]), which differ substantially from the more compact stances observed in other regions. Interestingly, intermediate niches such as [0.33, 0.33] capture transitional behaviors that combine elements of both crouching and leaning, showing that the algorithm does not merely populate extremes of the descriptor space but also uncovers meaningful intermediate variations. The qualitative differences between these behaviors confirm that the repertoire encodes a wide spectrum of distinct motion strategies, ranging from stable crouches to dynamic leaning postures. This visualization demonstrates that the method is not only able to optimize for reward but also produces an interpretable diversity of humanoid movements, with each descriptor niche corresponding to a clearly identifiable behavioral style.

5.2.3. Ablation Study

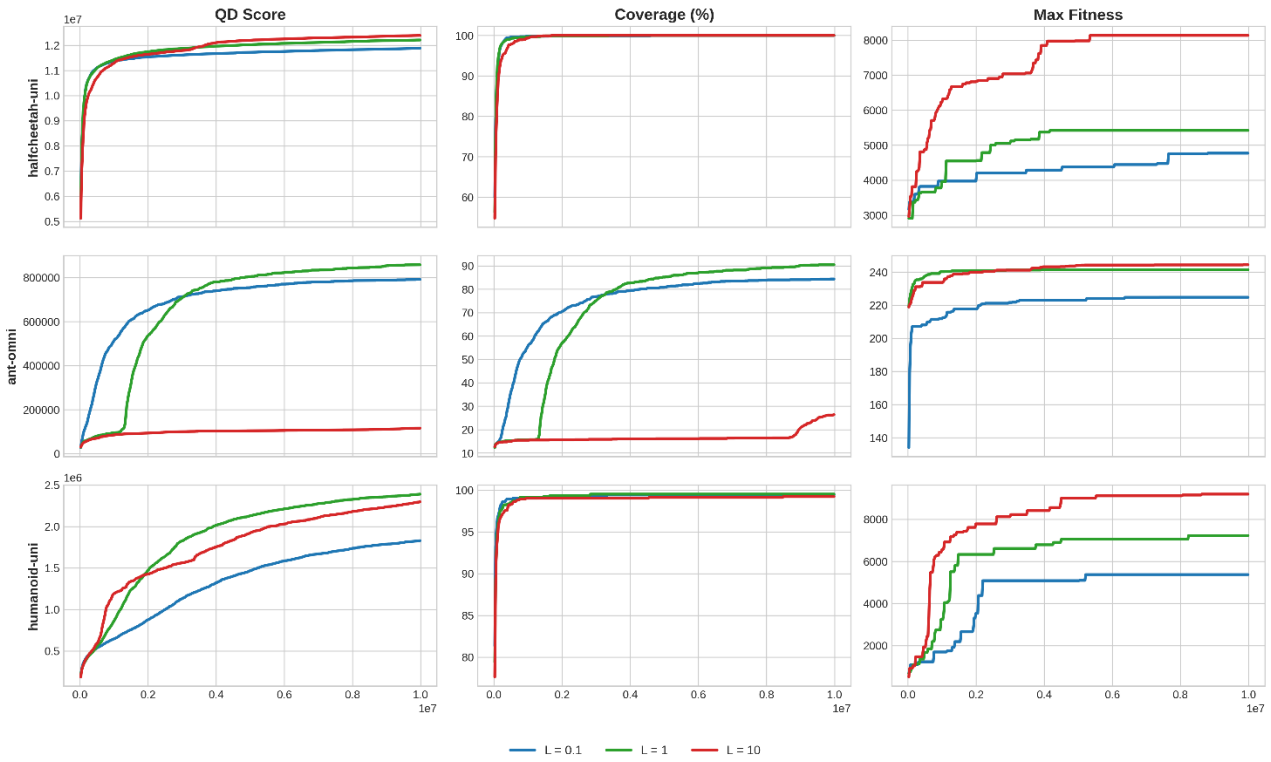


Figure 8: *Learning curves of QD Score, Coverage and Max fitness score across number of evaluations from different length scale values on three environments.*

Different length scales in similarity function.

To better understand the role of the descriptor-conditioned critic, we perform an ablation study on the length scale parameter L, which determines the smoothness of the similarity function between descriptors. A small value of L enforces strict separation between descriptors, while a larger value smooths the similarity landscape and allows reward information to generalize across a broader region. We evaluate three settings, L = 0.1, L = 1, and L = 10, on HalfCheetah-Uni, Ant-Omni, and Humanoid-Uni and show their resutls in Figure 8.

On Halfcheetah-Uni enviroments, the algorithms quickly achieve near-complete coverage across all three settings, indicating that exploration in this relatively simple task is not strongly affected by the length scale. However,

solution quality differs. Both L = 1 and L = 10 achieve higher QD Scores and Max Fitness than L = 0.1. The poor performance of the smallest scale suggests that overly strict descriptor matching prevents the critic from sharing reward information across nearby descriptors, limiting effective optimization.

On Ant-Omni task, the impact of L is more pronounced. Here, L = 0.1 and L = 1 achieves the best overall performance, rapidly increases coverage with high QD Score, reflecting its ability to well-generalize across descriptors. L = 10 performs worse, with lower coverage and diversity, indicating that excessive smoothing causes the critic to blur distinctions between behaviors, reducing the incentive to explore different niches.

In the high-dimensional Humanoid-Uni environment, both L = 1 and L = 10 outperform L = 0.1 by a wide margin. Interestingly, L = 10 yields the highest Max Fitness, showing that smoother similarity is advantageous in tasks where the search space is large and reward signals are sparse. However, L = 1 remains competitive, offering higher QD Score with slightly lower Max Fitness but more consistent coverage.

From these results, we summarize an intuitive way of thinking to effectively tune this sensitive hyperparameter. Halfcheetah-Uni and Humanoid-Uni tasks are pretty easy to expand diversity but extremely hard (especially Humanoid) to increase the fitness score, the Ant-Omni is more difficult to push diversity but relatively easy to achieve Max fitness score (by doing nothing). Therefore, if we desire to boost the quality, we should increase the L value to loosen the requirement of the similarity with the target descriptors. On the other hand, if we want to find more behaviors, the L value should be decreased to strictly enforce diversity.

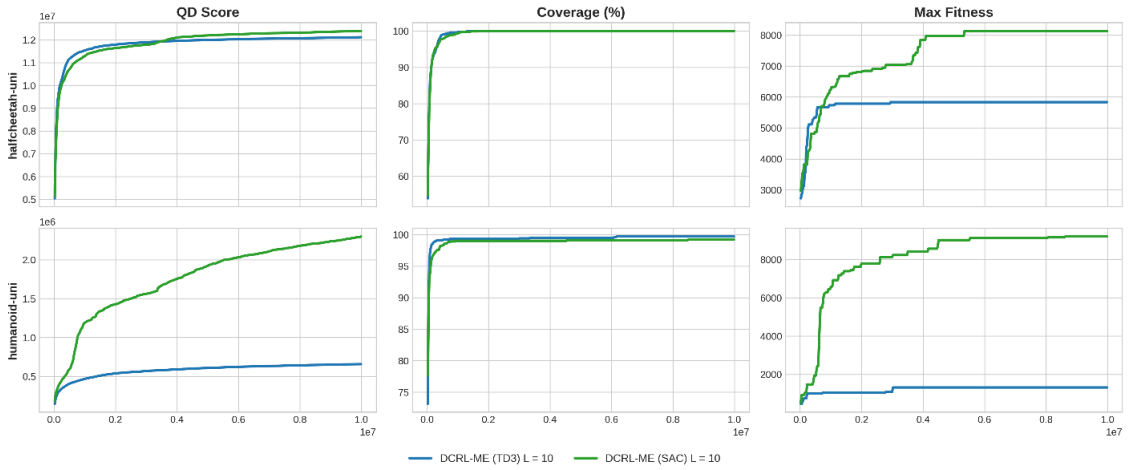


Figure 9: *Learning curves of QD Score, Coverage and Max fitness score across number of evaluations from TD3 and SAC variants for DCRL-ME with L = 10 on Halfcheetah-Uni and Humanoid-Uni.*

Sanity check of the original DCRL-ME framework.

Since our implementation employs different length scale values than the original work, a natural question arises: could the TD3-based variant of DCRL-ME achieve comparable or even superior performance if evaluated under the same length scale settings as our SAC-based method? To address this, we conduct a direct comparison between DCRL-ME (TD3) and DCRL-ME (SAC) on the two environments where the length scale differs, HalfCheetah-Uni and Ant-Omni, and show the results in Figure 9.

The results show that simply adjusting the length scale does not resolve the limitations of the TD3 backbone. In the challenging Humanoid-Uni task, DCRL-ME (TD3) continues to underperform, with both QD Score and Max Fitness remaining significantly lower than those of DCRL-ME (SAC), even after matching the length scale values. Similarly, in HalfCheetah-Uni, setting L = 10 fails to yield any meaningful improvement in fitness for the TD3-based variant. These findings suggest that the performance gap is not primarily due to the choice of length scale but rather stems from the inherent weaknesses of TD3 in scaling to high-dimensional continuous control, further reinforcing the advantage of adopting SAC as the RL backbone for descriptor-conditioned QD.

VI. CONCLUSIONS

In this work, we addressed the limitations of existing DCRL-ME methods in high-dimensional continuous control by integrating Soft Actor-Critic (SAC) into the descriptor-conditioned MAP-Elites framework. Our proposed approach, DCSAC-ME, reformulates the descriptor-conditioned policy as a stochastic Gaussian policy and using entropy-regularized RL methods as effective actor-critic training and variation operators within QD frameworks. Furthermore, by examining various length scale values, we extract some meaningful intuition of SAC's sensitivity for this hyperparameter and provide a way to adjust this value correctly. Through experiments on three Brax benchmarks, we demonstrated that DCSAC-ME not only preserves broad behavioral diversity but also substantially improves solution quality compared to its TD3-based counterpart. Our findings highlight the potential of entropy-regularized RL methods as effective combination with QD frameworks. Future work may explore extending DCSAC-ME to multi-task or transfer learning settings, where diverse skill archives can be composed hierarchically to tackle unforeseen challenges.

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